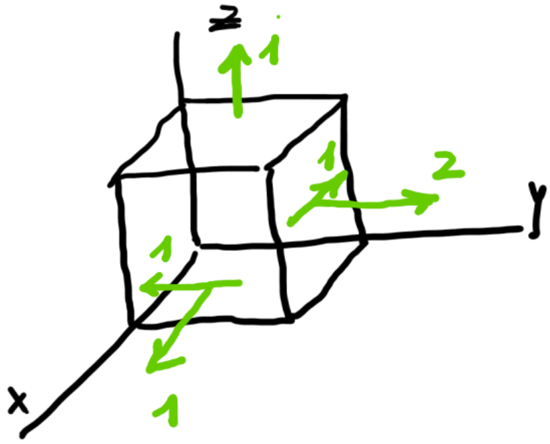
Esempio Analisi Stato di Tensione

Assegnato $\quad \underline{T} = \left| \begin{matrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{matrix} \right| \ MPa \qquad 1MPa = 1 \frac{N}{mm^2}$

1) Rappresentare $\underline{T}$

$$\underline{t}_x = \left| \begin{matrix} 1 \\ -1 \\ 0 \end{matrix} \right|, \quad \underline{t}_y = \left| \begin{matrix} -1 \\ 2 \\ 0 \end{matrix} \right|, \quad \underline{t}_z = \left| \begin{matrix} 0 \\ 0 \\ 1 \end{matrix} \right|$$



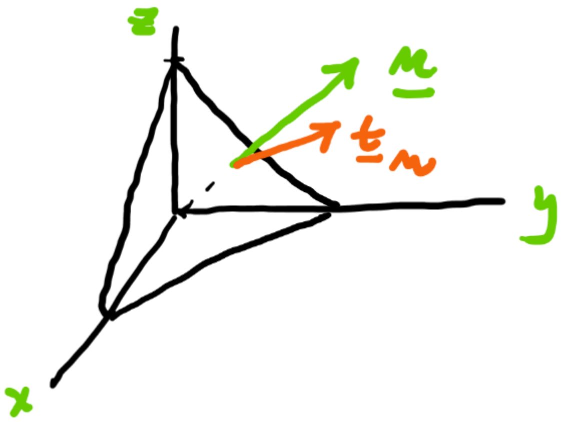
$\sigma_x = 1, \ \sigma_y = 2, \ \sigma_z = 1$

$\tau_{xy} = \tau_{yx} = -1$

$\tau_{zx} = \tau_{zy} = 0$

2) Calcolare $\underline{t}_n$ sul piano ottaedrico di normale $\underline{n} = 1/\sqrt{3} \left| \begin{matrix} 1 \\ 1 \\ 1 \end{matrix} \right|$

Cauchy: $\quad \underline{t}_n = \underline{T}\,\underline{n} = \frac{1}{\sqrt{3}} \left| \begin{matrix} 0 \\ 1 \\ 1 \end{matrix} \right| \ MPa$



$\underline{t}_n \ // \ yz$

$\underline{t}_n \ //$ bisettrice (yz)

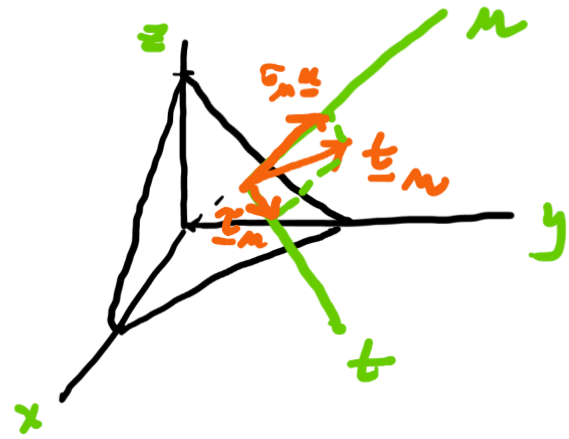
3) Effettuo la decomposizione

$$\underline{t}_n = \underline{\tau}_n + \sigma_n \underline{n}$$

Calcolo σ_n: $\sigma_n = \underline{n}^T \underline{t}_n = \frac{1}{\sqrt{3}} \begin{vmatrix} 1 & 1 & 1 \end{vmatrix} \cdot \frac{1}{\sqrt{3}} \begin{vmatrix} 0 \\ 1 \\ 1 \end{vmatrix} = \frac{2}{3}$ MPa

Calcolo $\underline{\tau}_n$: $\underline{\tau}_n = \underline{t}_n - \sigma_n \underline{n} = \frac{1}{\sqrt{3}} \begin{vmatrix} 0 \\ 1 \\ 1 \end{vmatrix} - \frac{2}{3} \cdot \frac{1}{\sqrt{3}} \begin{vmatrix} 1 \\ 1 \\ 1 \end{vmatrix}$

$$\underline{\tau}_n = \frac{1}{\sqrt{3}} \begin{vmatrix} -2 \\ 1 \\ 1 \end{vmatrix}$$



$$\underline{t}_n = \sigma_n \underline{n} + \underline{\tau}_n$$

4) Calcolare le tensioni e le direzioni principali

$$I_1 = tr(\underline{T}) = 4$$

$$I_2 = \begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = 4$$

$$I_3 = det(\underline{T}) = 1$$

Eq. Caratteristica: $\sigma_n^3 - 4\sigma_n^2 + 4\sigma_n - 1 = 0$

Tensioni Principali: $\sigma_1 = 1$, $\sigma_2 = \frac{3}{2} - \frac{\sqrt{5}}{2}$, $\sigma_3 = \frac{3}{2} + \frac{\sqrt{5}}{2}$

Direzioni Principali:

$$\begin{cases} \underline{T}\,\underline{n}_i = \sigma_i\,\underline{n}_i \\ \underline{n}_i^T\underline{n}_i = 1 \end{cases} \qquad \underline{n}_i = \begin{vmatrix} \alpha_i \\ \beta_i \\ \gamma_i \end{vmatrix}$$

Es: $\sigma_1 = 1$

$$\left(\underline{T} - \sigma_1\underline{I}\right)\underline{n}_1 = \underline{0}$$

$$\begin{vmatrix} 0 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{vmatrix}\begin{vmatrix} \alpha_i \\ \beta_i \\ \gamma_i \end{vmatrix} = \underline{0} \qquad \begin{cases} -\beta_i = 0 \\ \beta_i - \alpha_i = 0 \end{cases}$$

$\underline{n}_i^T\underline{n}_i = 1$: $\alpha_i^2 + \beta_i^2 + \gamma_i^2 = 1$

$$\alpha_i = \beta_i = 0$$

$$\underline{n}_1 = \pm\,\underline{e}_z = \pm\begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix} \qquad \gamma_i = \pm 1$$

Alle radici $\sigma_2 = \frac{3}{2} + \frac{\sqrt{5}}{2}$ e $\sigma_3 = \frac{3}{2} - \frac{\sqrt{5}}{2}$

sono associate le direzioni

$$\underline{n}_2 = \begin{vmatrix} -\frac{2}{\sqrt{5}+1} \\ 1 \\ 0 \end{vmatrix} \quad \text{ed} \quad \underline{n}_3 = \begin{vmatrix} \frac{2}{\sqrt{5}-1} \\ 1 \\ 0 \end{vmatrix}$$

5) Decomposizione $\underline{T} = \underline{S} + \underline{T}_d$

$$\underline{S} = \sigma_m \underline{I} = \frac{tr(\underline{T})}{3} \underline{I} = \frac{4}{3} \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \text{ MPa}$$

$$\underline{T}_d = \underline{T} - \underline{S} = \begin{vmatrix} -\frac{1}{3} & -1 & 0 \\ -1 & \frac{2}{3} & 0 \\ 0 & 0 & -\frac{1}{3} \end{vmatrix} \text{ MPa}$$

nota: $tr(\underline{T}_d) = 0$